Department of Information Technology

Course Code No: 23CS4T01

H.T.No: | | | | | | | | | | |

VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

II B. Tech II Semester (R23) Regular Examinations, April / May – 2025

OPERATING SYSTEMS

(CSE, IT, AI &DS & CSBS)

Time: 3hours Max. Marks: 70M

Note:1. Question Paper consists of two parts (Part-A and Part-B)

2. Answer all the questions in Part-A

3. Answer either a or b questions from Question number 2 to 6 in Part-B

PART –A

1	I	What is the role of system calls in an operating system?	L1	CO1	[2M
	II	List any two types of operating system structures.	L1	CO1	[2M
	III	Define multithreading and its benefits.	L1	CO2	[2M
	IV	Mention any two CPU scheduling algorithms.	L1	CO2	[2M
	V	State Peterson's solution and its purpose.	L1	CO3	[2M
	VI	What is the difference between deadlock prevention and avoidance?	L2	CO3	[2M
	VII	Define paging and explain its importance.	L1	CO4	[2M
	VIII	What is demand paging?	L2	CO4	[2M
	IX	List the different file access methods.	L1	CO5	[2M
	X	What are protection rings in operating systems?	L1	CO5	[2M

PART –B

UNIT – I

2(a)	I	Explain the services provided by an Operating System.	L1	CO1	[5M]
	II	Describe the different types of system calls with examples.	L1	CO1	[5M]

OR

2(b)	I	Discuss the structure of an operating system with a neat diagram.	L2	CO1	[5M]
	II	Describe how operating systems are booted and debugged.	L1	CO1	[5M]

UNIT – II

3(a)	I	Describe the process states and operations on processes.	L2	CO2	[6M]
	II	Explain the different multithreading models.	L1	CO2	[4M]
		OR			
3(b)	I	*Explain the various CPU scheduling algorithms with examples.*	L2	CO2	[6M]
	II	Compare preemptive and non-preemptive scheduling.	L2	CO2	[4M]
UNIT – III					
4(a)	I	Explain the classic problems of synchronization.	L2	CO3	[6M]
	II	*Describe the working of semaphores and monitors.*	L1	CO3	[4M]
		OR			
4(b)	I	*Explain the conditions that lead to deadlock with examples.*	L2	CO3	[5M]
	II	*Describe deadlock prevention and detection techniques.*	L2	CO3	[5M]
UNIT – IV					
5(a)	I	Discuss contiguous memory allocation and paging in detail.	L1	CO4	[5M]
	II	Explain the structure of page tables and the concept of swapping.	L1	CO4	[5M]
		OR			
5(b)	I	Explain demand paging and page replacement algorithms.	L1	CO4	[5M]
	II	What is thrashing? How can it be controlled?	L1	CO4	[5M]
UNIT – V					
6 (a)		Describe the file system interface and file access methods.	L2	CO5	[10M]
		OR			
6(b)	I	Explain file mounting, file sharing, and partitioning.	L1	CO5	[5M]
	II	Describe protection mechanisms using access matrix and protection rings.	L1	CO5	[5M]

*****